AI Project: Deliverable 2 Guide

Model Architecture & Empirical Evaluation

Deadline: 10th November, 2026

Objective

This deliverable shifts the focus from data foundations to the core ”intelligence” of your project. You are required to implement your model architecture (Phase 3) and conduct a rigorous empirical evaluation (Phase 4). Furthermore, this milestone strictly evaluates your ability to collaborate professionally using version control.

1 Phase 3: Model Architecture & Implementation

You must transition from data preprocessing to building the functional engine of your project.

- **Framework Selection:** Clearly implement your solution using industry-standard frameworks (*PyTorch, TensorFlow,* or *Scikit-Learn*).
- **Design:** Define the layers, loss functions, and optimizers.
- **Hyperparameters:** Document the tuning process. Provide a table of the parameters tested (Learning rate, Batch size, Dropout, etc.) and justify your final selections.

2 Phase 4: Training & Evaluation

A model is only as good as its validated performance.

- **Validation Strategy:** Implement a robust validation technique, such as K-Fold cross-validation or a clear Train/Validation/Test split (e.g., 70/15/15).
- **Metric Analysis:** Beyond simple accuracy, report Precision, Recall, F1-Score, and AUC-ROC. Justify why these metrics are appropriate for your specific problem (especially if dealing with imbalanced datasets).
- **Error Analysis & Bias Check:** Identify where the model fails. Does it struggle with specific classes or demographic subgroups? Include a ”Bias Check” section in your report.

3 Professional GitHub & Collaboration Policy

As students of the 5th semester, you are expected to operate at an industry-standard level regarding version control. Professionalism in Git is a core requirement of this deliverable.

No Manual Uploads: It is strictly prohibited to use the "Upload Files" button on the GitHub web interface. You must use the Git Command Line (CLI) or a Desktop client to add, commit, push, and sync changes. This is essential for effortless and professional collaboration.

Individual Coding Contribution: Every group member is required to contribute code to the project. I will evaluate individual participation by inspecting the **Git Commit History**. The logs must clearly show which student implemented which specific module, function, or preprocessing step. Each member must record their own work frequently; if a student's coding contribution is not visible in the Git history, they will receive a zero for the technical implementation phase.

4 Submission Checklist

Ensure the following are submitted:

1. **GitHub Repository:** Updated code in /src or /models folders. A requirements.txt file must be present to ensure environment reproducibility.

2. **Model Weights:** Do not push large binary files (>100MB) directly to Git. Provide a link in your README.md to an external host (e.g., Google Drive or Hugging Face) where the weights can be downloaded.

3. **LaTeX Report:** An updated PDF covering Phase 3 and 4. Include architecture diagrams, loss/accuracy curves, and a **"Contribution Table"** that maps Git commit authors to specific project modules.